Python Modules –

What are Modules?

- A **module** = a `.py` file containing Python code (functions, variables, classes).
- Purpose: **organize & reuse code** across projects.
- Analogy: toolbox → each module = a tool (e.g., `math`, `os`).

Creating & Using Modules

1. Create `my_module.py` → define functions/variables.
2. Import in notebook/script:
3. `import my_module`
4. `my_module.greeting("Alice")`
5. Example modules:
   - `math_utils.py`: `square(x)`, `cube(x)`, `factorial(n)`
   - `string_utils.py`: `reverse_string(s)`, `count_vowels(s)`

Import Variations

- **Specific function**:
- `from math_operations import add`
- **Multiple functions**:
- `from math_operations import add, subtract`
- **All functions**:
- `from math_utils import *`

Renaming Modules (Aliases)

- Syntax: `import module as alias`
- Example:
- `import calculator as calc`
- `print(calc.add(2,3))`
- Benefits: shorter names, avoid conflicts, improve readability.

Built-in Modules

1. platform

- System info: `platform.system()`, `platform.release()`, `platform.version()`
- Python info: `python_version()`, `python_compiler()`

2. datetime

- Current time: `datetime.now()`
- Formatting: `strftime("%Y-%m-%d")`
- Parsing: `strptime("2023-06-01", "%Y-%m-%d")`
- Arithmetic: `timedelta(days=1)`

3. math

- Arithmetic: `sqrt()`, `ceil()`, `floor()`, `fabs()`, `factorial()`
- Trigonometry: `sin()`, `cos()`, `tan()`, constants `pi`, `e`
- Logs/exp: `log()`, `log10()`, `exp()`
- Conversion: `degrees()`, `radians()`

4. random

- Numbers: `random()`, `randint(a,b)`, `uniform(a,b)`
- Choices: `choice(seq)`, `sample(seq,k)`
- Shuffle: `shuffle(list)`
- Seed: `seed(42)` → reproducibility

5. os

- File system: `getcwd()`, `listdir()`, `mkdir()`, `remove()`, `rmdir()`, `rename()`
- OS type: `os.name` → `'posix'` (Linux/macOS), `'nt'` (Windows)

File Handling

- Open file: `open(filename, mode)`
  - Modes: `'r'` (read), `'w'` (write), `'a'` (append), `'x'` (create)
- Best practice:
- `with open("file.txt", "r") as f:`
- `    data = f.read()`
- Ensures file closes automatically.
